

8.1. For matrix A, B det. axis of rotation and rotation angle.

Axis of rotation = line passing through the origin in the direction of the eigenvectors for eigenvalue 1.

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \Rightarrow \text{Tr } A = -\frac{1}{3} - \frac{2}{3} + \frac{2}{3} = -\frac{1}{3}$$

$$B = \frac{1}{11} \begin{bmatrix} -9 & -2 & 6 \\ 6 & -6 & 7 \\ 2 & 9 & 6 \end{bmatrix}$$

$$\frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \frac{1}{3} \begin{bmatrix} -1-1 & 2-1 & -2-1 \\ -2-1 & -2-1 & -1-1 \\ -2-1 & 1-1 & 2-1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad / \cdot 3$$

$$\Rightarrow \begin{bmatrix} -2 & 1 & -3 \\ -3 & -3 & -2 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -4 & -1 & -5 \\ -5 & -5 & -4 \\ -5 & -2 & -1 \end{bmatrix} \dots ?$$

$$\Rightarrow \begin{bmatrix} -4 & 2 & -2 \\ -2 & -5 & -1 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 1 \\ -2 & -5 & -1 \\ -2 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 1 \\ 0 & -6 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} 2x - y + z = 0 \\ -6y = 0 \Rightarrow y = 0 \end{cases} \Rightarrow \begin{cases} 2x + z = 0 \\ z = -2x \end{cases} \Rightarrow y = 0$$

$\Rightarrow$ eigenspace for $\lambda=1$: $V_1 = \{(t, 0, -2t) \mid t \in \mathbb{R}\} =$ line passing through the origin = rotation axis $\Rightarrow$ eigenvectors: $\neq 0$ vectors in V_1 .

λ eigenvalue

$\Downarrow$

$(A - \lambda I)v = 0 \Rightarrow$ eigenvectors

$\Rightarrow$ solutions $\Rightarrow$ subspace

$\Rightarrow$ eigenspace $\Rightarrow$ no eigenvect. in the same direction

Angle of rotation Θ : $\boxed{\text{tr}(A) = 1 + 2\cos\Theta}$

$$\Rightarrow \cos\Theta = \frac{\text{tr}(A) - 1}{2} = \frac{-\frac{1}{3} - 1}{2} = -\frac{2}{3}$$

$$\Rightarrow \Theta = \arccos\left(-\frac{2}{3}\right)$$

B $\Rightarrow$ similar.

8.2. Show that the transformation is a glide-rotation.

Rotation angle = ? Rotation axis = ?

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \mapsto \frac{1}{45} \begin{bmatrix} 37 & -20 & -16 & 15 \\ 16 & 40 & -13 & -30 \\ 20 & 5 & 40 & 30 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

(K1) $\det A = \frac{1}{45} \begin{bmatrix} 37 & -20 & -16 \\ 16 & 40 & -13 \\ 20 & 5 & 40 \end{bmatrix}$ matrix of the linear part.

• $\begin{cases} A \cdot A^T = I_3 \\ \det A = 1 \end{cases} \Rightarrow A \text{ is in } SO(3) \Rightarrow \text{affine space} = \text{direct isometry}$
(matrix)

• $A \neq I_3 \Rightarrow$ the map is not identity or translation.

• Direct isometry classification $\Rightarrow$ rotation / glide rotation.

$\downarrow$
fixes the
rotation
axis

$\downarrow$
does not fix any point

• Investigate fixed points of the map :

$$\frac{1}{45} \begin{bmatrix} 37 & -20 & -16 \\ 16 & 40 & -13 \\ 20 & 5 & 40 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \frac{1}{45} \begin{bmatrix} 15 \\ -30 \\ 30 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Rightarrow \left(\frac{1}{45} A - I_3 \right) \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \frac{1}{45} b = 0 \quad | \cdot 45$$

$$\Rightarrow (A - 45I_3) \begin{bmatrix} x \\ y \\ z \end{bmatrix} + b = 0$$

Subtract $\leftarrow$
45 from
diagonal positions

$$\Rightarrow \begin{bmatrix} -8 & -20 & -16 \\ 16 & -5 & -13 \\ 20 & 5 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 15 \\ -30 \\ 30 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$A =$ linear part = rotation in $\mathbb{R}^3$ $\rightarrow$ eigenvalue = 1 $\Rightarrow$ direction of axis (2)
 $\rightarrow \dim(\text{Ker}(A - I))$

$$\text{rank}(A - I) + \text{nullity}(A - I) = 3$$

$$\text{rank}(A - I) = 3 - 1 = 2. \quad (\text{rotation always fixes an axis (line)})$$

- Check rank of extended matrix. $\begin{matrix} \nearrow 2 \Rightarrow \text{rotation} \Rightarrow \exists \text{ fixed points} \\ \searrow 3 \Rightarrow \text{glide rotation} \Rightarrow \nexists \text{ fixed points} \end{matrix}$
 $\Rightarrow$ calculation

$$\begin{bmatrix} -5 & -20 & -16 & -15 \\ 16 & -5 & -13 & 30 \\ 20 & 9 & -5 & -30 \end{bmatrix} \sim \dots \begin{bmatrix} 1 & 0 & -\frac{6}{11} & -\frac{75}{11} \\ 0 & 1 & \frac{103}{110} & \frac{27}{11} \\ 0 & 0 & 0 & \frac{15}{2} \end{bmatrix}$$

no fixed points $\Leftarrow$ no solutions $\Leftarrow$ inconsistent

- H_2 : $\Rightarrow$ calculate rotation axis for lin. repres. by A .
 $\rightarrow$ check if translation part of the map

$$\frac{1}{45} \begin{bmatrix} 15 \\ -30 \\ 30 \end{bmatrix} \text{ is orthogonal to the axis}$$

$\rightarrow$ if yes $\Rightarrow$ rotation

$\rightarrow$ if not $\Rightarrow$ glide-rotation

$$\text{rank}(A - 45I_3) = 2$$

$$\text{rank}[A - 45I_3 | b] = 3$$

Q.3. Show that the transformation is a glide-reflection.

Reflecting plane?

Translation vector?

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \mapsto \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 & 4 \\ -2 & -1 & 2 & 2 \\ 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad (\text{map})$$

$$A = \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 \\ -2 & -1 & 2 \\ 1 & 2 & 2 \end{bmatrix} \text{ matrix of the linear part of the map.}$$

$$A \cdot A^T = I_3 \Rightarrow A \in O(3)$$

$$\det A = -1 \Rightarrow \text{indirect isometry.}$$

$$\Phi(P) = AP + v, \quad v = \frac{2}{3} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

reflection

glide reflection

rotation-reflection

• Investigate linear part of the map:

$$\psi(P) = AP$$

$A \in O(3) \setminus SO(3) \Rightarrow -1$ eigenvalue

$\Rightarrow$ eigenvectors for -1 are normal vectors for plane which are invariant under ψ

• eigenvectors for -1 : $A v = \lambda v$, $\lambda = -1$:

$$A v = -v \Rightarrow \frac{1}{3} \begin{bmatrix} 5 & -2 & 1 \\ -2 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 5 & -2 & 1 \\ -2 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \sim \begin{bmatrix} 0 & -12 & -24 \\ 0 & 6 & 12 \\ 1 & 2 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 5 \\ 0 & 6 & 12 \\ 0 & -12 & -24 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 5 \\ 0 & -12 & -24 \\ 0 & 6 & 12 \end{bmatrix} \sim$$

$$\sim \begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x+z=0 \\ y+2z=0 \end{cases} \Rightarrow \begin{cases} 2x+2z=0 \Rightarrow x \\ y+2z=0 \end{cases}$$

$$\Rightarrow \begin{cases} x = -z \\ y = -2z \\ z = z \end{cases} \Rightarrow \text{eigenvectors are spanned by:}$$

$$w = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$2x - y = 0$$

$$y = 2x = -2z$$

• The other 2 eigenvalues for ψ are either

both 1 $\Rightarrow \psi = \text{reflection}$

both complex $\Rightarrow \psi = \text{rotation reflection}$

• find eigenvalues $\Rightarrow$ characteristic polynomial:

$$\det(A - \lambda I) = \det \frac{1}{3} \begin{bmatrix} 2-3\lambda & -2 & 1 \\ -2 & -1-3\lambda & 2 \\ 1 & 2 & 2-3\lambda \end{bmatrix} = -(\lambda-1)^2(\lambda+1)$$

$$\Rightarrow \lambda_1 = \lambda_2 = 1$$

$$\Rightarrow \lambda_3 = -1$$

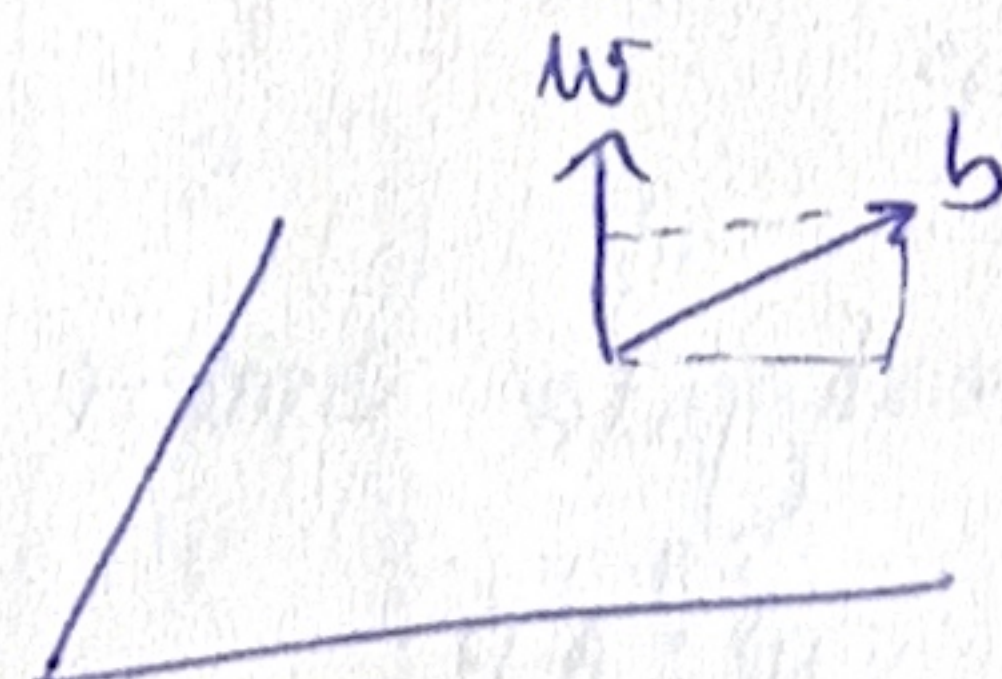
$\Rightarrow$ admits 1 as eigenvalue $\Rightarrow \psi$ reflection.

• ψ is the reflection ψ , followed by translation T_v where $v = \frac{2}{3} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$.

• lecture $\Rightarrow$ reflection plane:

$$\bar{u}: \langle w, x \rangle = \frac{1}{2} \langle w, v \rangle$$

$$(z) \bar{u}: x + 2y - z = 1$$



Moreover, the translation vector is the component of b which is $\perp \tilde{w}$

$$v - \text{Pr}_{\tilde{w}}(v) = v - \frac{\langle w, v \rangle}{\langle w, w \rangle} w = \frac{2}{3} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{6} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

8.4 Dim 3, show that the composition of a rotation with 60° around the x-axis, followed by a rotation with 30° in the z axis is a rotation. Rot. angle = ?

• Rot. with 60° around Ox: $R_1 = \begin{bmatrix} \textcircled{1} & 0 & 0 \\ 0 & 1/2 & -\sqrt{3}/2 \\ 0 & \sqrt{3}/2 & 1/2 \end{bmatrix} \Rightarrow \det R_1 = 1$

• Rot. with 30° around Oz: $R_2 = \begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 \\ 1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & \textcircled{1} \end{bmatrix} \Rightarrow \det R_2 = 1.$

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

R_1, R_2 have $\det = 1$.

$\Rightarrow \det(R_2 R_1) = 1 \Rightarrow$ composition is direct isometry.

$\left. \begin{array}{l} R_1 \text{ fixes the origin} \\ R_2 \text{ fixes the origin} \end{array} \right\} \Rightarrow$ composition fixes the origin. $\Rightarrow$ rotation. (a point)

Trace-formula for finding rot. angle:

$$R = R_2 R_1 = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{1}{2} & \frac{\sqrt{3}}{4} & -\frac{3}{4} \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \Rightarrow \text{Tr}(R) = \frac{3\sqrt{3}+2}{4}$$

$$\Rightarrow \boxed{\cos(\theta) = \frac{\text{Tr}(R) - 1}{2}} = \frac{3\sqrt{3} - 2}{8}$$

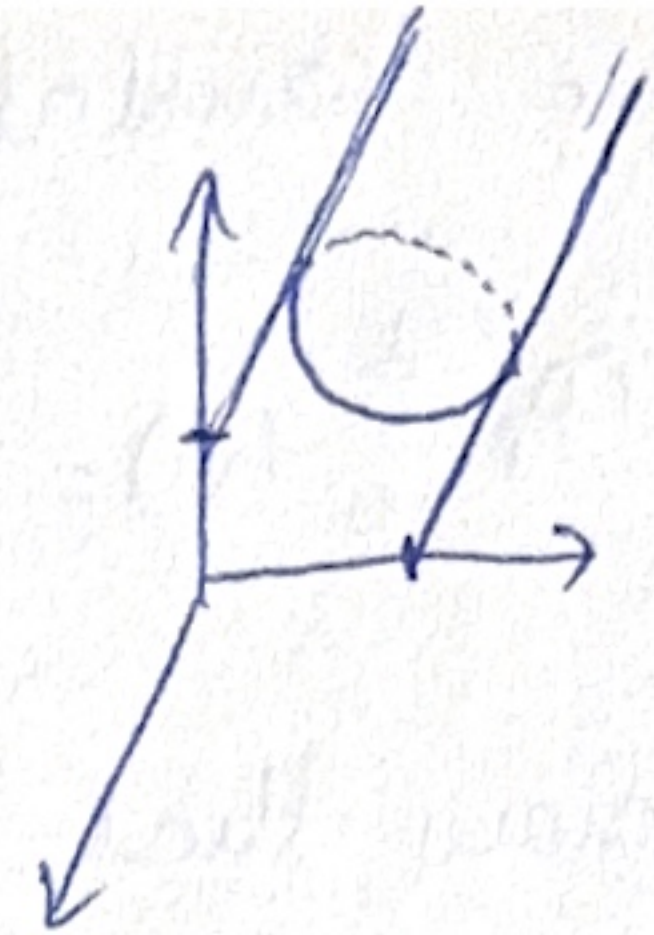
$$\Rightarrow \theta \approx 66.45^\circ$$

8.5. Using rotations around the coord. axes, give a parametrization of a cylinder with axis $\mathbb{R}(0,1,1)$ and diameter $\sqrt{2}$.

The cylinder contains the line: $\ell: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

... and has axis: $\ell': \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

We may construct a cylinder $\tilde{\mathcal{C}}$ of diam $\sqrt{2}$ and axis Oz and then rotate it in the right position.



$$\tilde{\mathcal{C}} = \tilde{\ell} : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\tilde{\mathcal{C}} = \text{Rot}_{K, \theta}(\tilde{\ell}) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{\sqrt{2}}{2} \\ t \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} \sin \theta \\ \frac{\sqrt{2}}{2} \cos \theta \\ t \end{bmatrix}$$

$$\Rightarrow \mathcal{C} = \text{Rot}_{I, -45^\circ}(\tilde{\mathcal{C}}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2}/2 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} -\frac{\sqrt{2}}{2} \sin \theta \\ \frac{\sqrt{2}}{2} \cos \theta \\ t \end{bmatrix} \dots$$